

DIP-CSE340 ASSIGNMENT 3
Deadline : 10/30/2024

Total Marks : 40 Marks

Total Days : 14 Days

Instructions:

- ◆ Standard institute plagiarism policy holds.
- ◆ Even a second late will be considered late, so do submit your assignments at least before 11:50 PM on the deadline date to avoid any unexpected issues.
- ◆ The Kaggle competition will close at 11:55 PM on the date of deadline.
- ◆ Submit files : *report_Rollnumber.pdf* and Jupiter notebook *code_Roll_Number.ipny*.
- ◆ Your code and report with the images generated will be checked before you are graded.
- ◆ Uploading the original images will result in an error. So if you can achieve a 0 score, mail regarding the same.

Section A

20 marks

Given the two images and their noisy version with distortions, restore them back to their original form

1. Consider the image (b) in Figure 1, and perform the operations below on the image (b) to restore the image in (a).
 - (a) Denoise the image
 - (b) Inpaint to remove the patches and text

The mask for inpainting is not provided; therefore, you need to approximate the same from the image. (Use the image processing techniques you have learned)

2. Write a single function/class/ module that takes an image and performs the necessary operation to give the output. The function should be agnostic to similar images (i.e., you cannot use it if-and-else perform different operations on the images (Image1 and Image2)). Upload the code in the classroom as a Jupiter notebook *code_Roll_Number.ipny*. **A missing code file will result in zero marks in the assignment.**
3. Beat the inpainting score through image quality enhancement in the [Kaggle challenge](#). You are fighting against each one in the class. 10 marks
 - (a) The image in figure 1 (Image1) and a similar image (Image2) will be provided.



(a) Original image

(b) degraded image

Figure 1: The image (b) is the noisy degraded image of (a)

- (b) The metric used for the evaluation is the mean squared error (MSE) between the Restored Image (Restored Image1 and Restored Image2) and the original image.

$$average(Image1, Image2) = \frac{\sum_{i=1}^2 MSE(Image_i, RestoredImage_i)}{2}$$

- (c) The size of the image to be uploaded is 1200×1200 , and the format is .png
- (d) You are allowed to use any library, inbuilt function or code base but not deep learning.
- (e) Submissions allowed per day 30.
4. A tie in the competition will result in those with better presentation and techniques getting higher marks. For the same, the code and its time complexity will also be considered.

Section B

20 marks

Submit a latex report of 2 pages

1. Identify the type of noise in the images and explain the methods used for the image de-noising.
Note that different noises are added to Image1 and Image2. 4 marks
2. Write the pseudo-code of the algorithm used for inpainting and explain each and every step in the pipeline in detail. 8 marks
3. Demonstrate using images of your choice: 5 marks

- (a) Image 1: merit of your selected inpainting algorithm
 - (b) Image 2: demerit of your selected inpainting algorithm
4. Explain any additional method and/or pre/post-processing techniques you used to improve the quality of your output as you progressed in the challenge. 3 marks

END